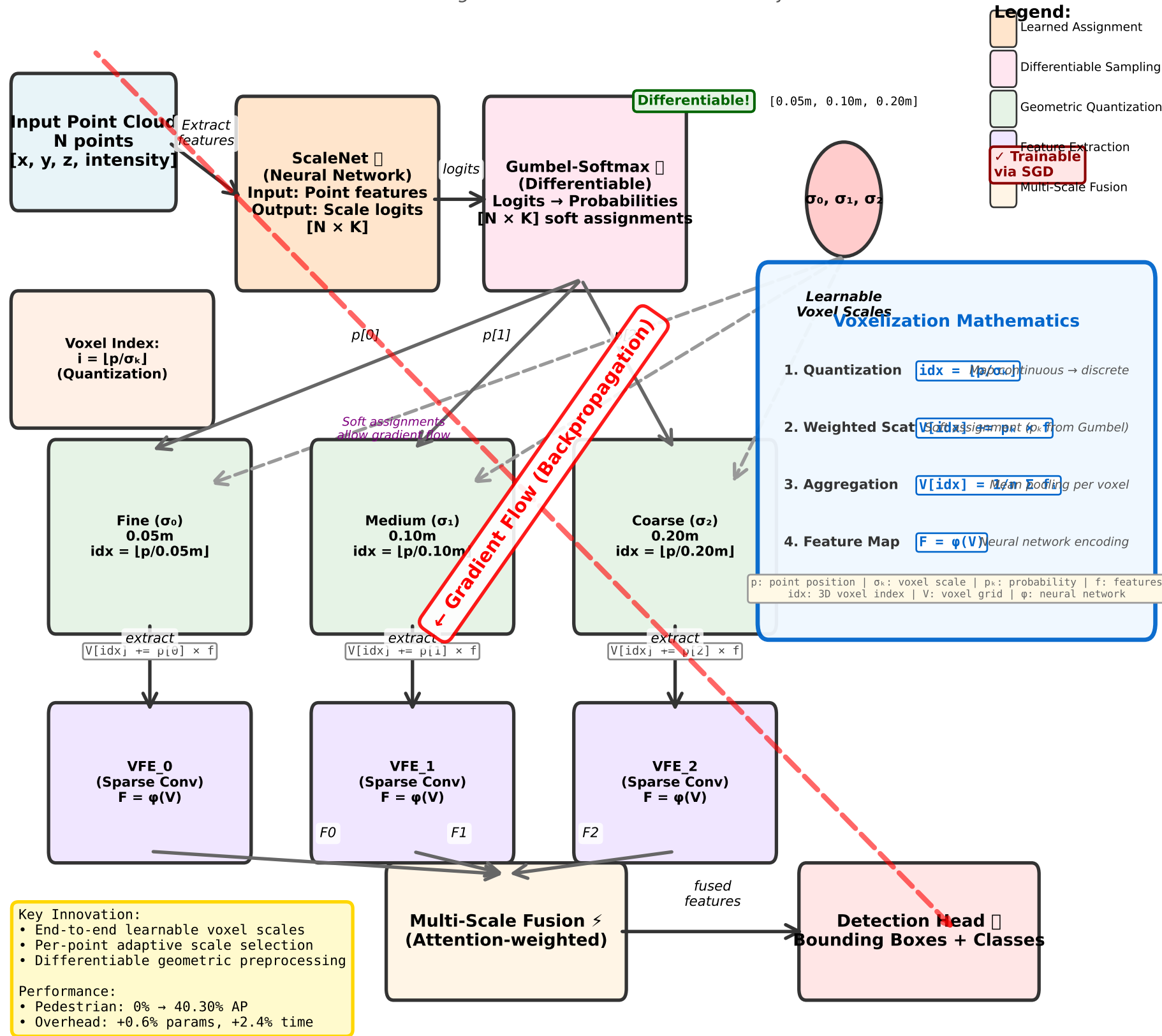


VoxAdapt Architecture: Learnable Multi-Scale Voxelization

End-to-end learnable geometric discretization for 3D object detection



VoxAdapt treats voxel scales ($\sigma_0, \sigma_1, \sigma_2$) as trainable parameters. ScaleNet learns to assign each point to the optimal scale, Gumbel-Softmax enables gradient flow through discrete choices, and multi-scale features are fused for final detection. All components are jointly optimized end-to-end.